

# SOPHIE SHIH

New York, NY | [sophie59595@gmail.com](mailto:sophie59595@gmail.com) | [LinkedIn](#)

Versatile, self-driven student with great communication skills and strong foundations in programming and medical AI / LLMs.

## EDUCATION

### Duke University

Expected: May 2030

Computer Science and Neuroscience

- Relevant Coursework: Data Structures & Algorithms
- Involvements: Catalyst Tech Society, Duke Chronicle, StreetMed Dance Team, Club Tennis

### Stuyvesant High School

GPA: 4.0/4.0 || 97

Jun. 2026

- Honors Classes: AP Computer Science, AP Physics C Mech, AP Physics C E&M, AP Calculus BC, AP Chemistry, AP Mandarin, AP US Government, AP Macroeconomics, AP European History

## EXPERIENCE

### Mass General Brigham

Jun. 2025 - Sep. 2025

Research Intern

New York, NY

- Independently developed an interactive, research-based app exploring links between the microbiome and Parkinson's disease, integrating AI agents, the MINERVA knowledge graph, semantic literature search, and personalized health insights.
- Designed the Streamlit interface and dynamic visualizations, conducted biomedical research, and authored a comprehensive [thesis](#) examining the use of AI to translate complex scientific findings into accessible educational tools.

### TheCoderSchool

Nov. 2024 - Jun. 2025

Intern / Code Coach

New York, NY

- Developed custom code projects and challenges to reinforce key programming concepts and foster creativity
- Designed and led coding projects that introduced fundamental physics concepts—such as gravity, acceleration, and collisions—through interactive games in Scratch

### Weill Cornell Radiology

Jul. 2024 - Jun. 2025

Research Intern

New York, NY

- Annotated MRIs of ADPKD (Autosomal Dominant Polycystic Kidney Disease) patients and helped to build an AI deep learning model with the ability to segment organs for MR abdomen and pelvis studies
- Project using few-shot LLM prompting to generate patient-friendly explanations for incidental findings on MRI ([RSNA abstract](#))
- Explored mathematical models and spatial algorithms underpinning image segmentation and neural network training for volumetric data

### MD.ai

Jun. 2023 - Jul. 2023

Intern

New York, NY

- Presented weekly regarding LLMs for healthcare use cases, curating high-quality medical information for question / answering
- Shadowed engineers and helped with testing of features in their MD.ai Reporting product including automated radiology report generation, which uses LLMs in a HIPAA-compliant environment
- Worked on a project using LLM Agents to generate patient-friendly letters (<https://arxiv.org/pdf/24me08.01112>), project winning 2nd place at the SIIM Hackathon ([slides](#) / [video](#))

### Believers

Jul. 2024, 2025

Volunteer

Tokyo, Japan

- Believers provides a solution to a growing issue of young students in Japan dropping out due to rigid school environments and lack of social support by creating a third environment/ middle-ground between home and school
- Worked collaboratively with elementary school students in various activities including crafts, geography, cooking, music, etc.

## AWARDS & CERTIFICATIONS

- SIIM (Society for Imaging Informatics in Medicine) Hackathon — 2nd Place, 2024 ([slides](#) / [video](#)) and 2025 ([slides](#) / [video](#))
- CITI Program Training: Biomedical Research Investigators and Key Personnel, IRB Members, Good Clinical Practice

## SKILLS & TOOLS

- **Languages:** Java, Python, HTML/CSS, JavaScript, NetLogo, Racket
- **Tools:** Git/GitHub, Notion, OnShape/CAD, Streamlit, Generative AI & LLMs, Docker
- **Skills:** Biomedical Research, LLM Prompt Engineering, Machine Learning, Scientific Writing